Software Requirements Specification

for

Research Paper PDF Extraction

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1. Introduction

1.1 Purpose

This document outlines the requirements for the project "Research Paper PDF Extraction to Excel," which aims to extract specific information from research papers in PDF format and save the extracted data into Excel files. The project will provide a web interface for easy file uploads and result downloads.

1.2 Document Conventions

This SRS follows the IEEE standard for software requirements specifications. Key terms are highlighted in bold. Requirements are prioritized as follows:

- **Headings:** Major sections are bold and use 16 pt font size; subheadings use 14 pt; text use 12 pt.
- **Bold Text:** Used for section titles and important terms.
- **Italics:** Used for emphasis and to introduce new terms.
- **Numbered Lists**: Used for sequential or prioritized items.

1.3 Scope

The application will allow users to upload multiple PDF files, extract relevant data (Title, Date, Emails, Journal Name, and Abstract), and download the results in Excel format. The frontend will be user-friendly, built using Bootstrap, while the backend will be powered by Flask and Python

2. Overall Description

2.1 Product Perspective

The system will consist of a web application that integrates a file upload mechanism with a backend processing to extract data from PDFs and a frontend for user interaction.

2.2 Product Functions

The major functions of the E-learning platform include:

- Upload PDFs via the web interface.
- Extract specific information from each PDF.
- Save extracted data into Excel files.
- Allow users to download the processed files.

2.3 User Classes and Characteristics

- Researchers or students needing to extract data from multiple research papers.
- Users with basic computer skills, familiar with web interfaces.

3. Functional Requirements

3.1 User Requirements

1. **PDF Upload**

   - Users must be able to upload multiple PDF files.
   - The system must support common PDF formats.

2. **Data Extraction**

   The system must extract the following:

   - **Title**: Extracted from the first page of the
   - **PDF Date:** Extracted using regex patterns.
   - **Emails**: Extracted using regex patterns.
   - **Journal Name**: Identified through keywords.
   - **Abstract Text**: Extracted text before the specified heading.

3. **Save to Excel**

   The system must generate two Excel files:

- **Text before Abstract**: Contains chunks of text extracted before the specified heading.
- **Main Data**: Contains Title, Journal Name, Date, and Emails.

4. **Download Functionality**

    - Users must be able to download the generated Excel files.

4. External Interface Requirements

4.1 User Interfaces

- **Upload Form:** A form where users can select PDF files and specify a heading to extract data.

- **Result Page:** A page displaying links to download the generated Excel files.

4.2 Hardware Interfaces

- No specific hardware interfaces are required, as the application will be web-based.

4.3 Software Interfaces

The application will use:

- **Flask**: For backend processing and routing.

- **Fitz (PyMuPDF):** For extracting text from PDFs.

- **Pandas**: For manipulating data and exporting it to Excel.

5. Use Cases

5.1 Upload PDFs

Actor: User
Precondition: User is on the upload page.

Main Flow:

- User selects one or more PDF files.
- User specifies the heading (default is "Abstract").
- User clicks "Upload."

5.2 Download Results

Actor: User
Precondition: Data extraction has been completed.

Main Flow:

- The user is directed to the results page.
- User clicks on download links for the generated Excel files.

6. Other Nonfunctional Requirements

6.1 Performance

- The system must handle multiple file uploads and process them within a reasonable time frame.

6.2 Usability

- The front end must be responsive and user-friendly, ensuring ease of navigation.

6.3 Security

- Uploaded files must be securely handled, and user data must not be stored unnecessarily

7. Input Format

- PDF Files: Users can upload one or more research papers in PDF format.
- Heading Input: A text input where the user specifies a heading (e.g., "Abstract").

8. Output Format

- **Text Before Abstract Excel File:**
  - Contains chunks of text extracted before the specified heading.
  - Each row represents a chunk of text.

- **Main Data Excel File:**
  - Contains extracted Title, Journal Name, Date, and Emails.
  - Columns: Title, Journal Phrase, First Date, Emails.

9. Example Input and Output Format

- **Example Input**

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MARGINAL PERCEPTRON FOR NON-LINEAR AND MULTI CLASS CLASSIFICATION

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ABSTRACT

Generalization error of classifier can be reduced by larger margin of separating hyperplane. The proposed classification algorithm implements margin in classical perceptron algorithm, to reduce generalized errors by maximizing margin of separating hyperplane. Algorithm uses the same updation rule with the perceptron, to converge in a finite number of updates to solutions, possessing any desirable fraction of the margin. This solution is again optimized to get maximum possible margin. The algorithm can process linear, non-linear and multi class problems. Experimental results place the proposed classifier equivalent to the support vector machine and even better in some cases. Some preliminary experimental results are briefly discussed.

KEYWORDS

Perceptron, Non-linear Classification, Multi Class Classification, Support Vector Machine & Generalization Error.

1. INTRODUCTION

A perceptron is a binary classifier, initially developed as a model of the biological neuron [1]. Internally, it computes a linear combination of real-valued labeled samples and predicts the classes for unlabeled samples. If two different set of samples can be separated by a straight hyperplane, they are called linearly separable. As a consequence, irrespective of the training algorithm used, linear classifiers like the perceptron cannot arrive at correct predictions for all potential instances unless the given problem is linearly separable. Classifiers generally face the problem of incorrect classification for those instances which are closer to separating hyperplane, known as generalization error. The number of generalization errors depends on the margin or distance between the positive and negative samples [2]. The hyperplane that maximizes the margin to the closest positive and negative sample is called the optimal hyperplane. The perceptron will not necessarily find the optimal hyperplane. However, the perceptron is an online learning classifier, means it process one sample at a time which allowed allows it to spare time and memory resources for handling large-scale classification problems.

- **Example Output**

Title	Journal Phrase	First Date	Emails
MARGINAL PERCEPTRON FOR NON-LINEAR AND MULTI CLASS CLASSIFICATION	Journal on Soft Computing	1, August 2012	1hemant061018@gmail.com, 2sgupta@sgsits.ac.in

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1	**Text Before Abstract**
2	International Journal on Soft Computing,
3	Artificial Intelligence and Applications (IJSCAI),
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